Plots with Feature Values

23/02/2024

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: install.packages("ggplot2").

```
library(stringr)
library(ggplot2)
library(ggrepel)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(ggExtra)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'

## The following object is masked from 'package:dplyr':
##
##     combine
```

```
library(ggpubr)
```

Load Data

Load csv file with quantitative feature values for sign sequences per object. Comparative sample: A comparative sample of feature values for languages across the world is dreived from the Text Data Diversity (TeDDi) sample (Moran et al 2022).

```
# SIGNBASE
# load signbase estimations for strings on objects
sign.data <- read.csv("~/Github/UpperPalCombinatorics/results/signBase_features.csv")
# select Aurignacian objects
sign.data <- sign.data[sign.data$archaeological_unit == "Aurignacien", ]

# select columns for direct comparison with TeDDi sample
sign.data.selected <- select(sign.data, c('archaeological_unit', 'num.char', 'huni.chars', 'hrate.chars
```

```
                                 'ttr.chars', 'rm.chars'))
# change column names
colnames(sign.data.selected) <- c('subcorpus','num.char','huni.chars', 'hrate.chars',
                         'ttr.chars', 'rm.chars')

# CDLI
# read feature values for proto-cuneiform from CDLI
# Uruk V
procuneiV.data <- read.csv("~/Github/UpperPalCombinatorics/results/procunei_urukV_features.csv")

# add column with subcorpus information
procuneiV.data$subcorpus <- rep("Proto-Cuneiform (Uruk V)", nrow(procuneiV.data))

# select columns for direct comparison with NaLaFi sample
procuneiV.data.selected <- select(procuneiV.data, c('subcorpus', 'num.char', 'huni.chars', 'hrate.chars
                                 'ttr.chars', 'rm.chars'))
# Uruk IV
procuneiIV.data <- read.csv("~/Github/UpperPalCombinatorics/results/procunei_urukIV_features.csv")

# add column with subcorpus information
procuneiIV.data$subcorpus <- rep("Proto-Cuneiform (Uruk IV)", nrow(procuneiIV.data))

# select columns for direct comparison with NaLaFi sample
procuneiIV.data.selected <- select(procuneiIV.data, c('subcorpus', 'num.char', 'huni.chars', 'hrate.cha
                                 'ttr.chars', 'rm.chars'))

# TeDDI
# read feature values for character sequences of the TeDDi sample
teddi.data <- read.csv("~/Github/UpperPalCombinatorics/results/teddi_features.csv")
# add column with subcorpus information
teddi.data$subcorpus <- rep("TeDDi", nrow(teddi.data))
# select columns
teddi.selected <- select(teddi.data, c('subcorpus','num.char','huni.chars', 'hrate.chars',
                           'ttr.chars', 'rm.chars'))
```

Preprocessing

```
# combine the data frames
estimations.df <- rbind(sign.data.selected, procuneiV.data.selected,
                        procuneiIV.data.selected, teddi.selected)
nrow(estimations.df)
```

```
## [1] 1934
```

```
# select sign strings of certain length
estimations.df <- estimations.df[estimations.df$num.char > 5,]

# select only certain subcorpora for plotting
estimations.df.selected <- estimations.df[estimations.df$subcorpus == "Aurignacien"|
                                          #estimations.df$subcorpus == "Aurignacien/Gravettian"|
                                          estimations.df$subcorpus == "Proto-Cuneiform (Uruk V)"|
                                          estimations.df$subcorpus == "Proto-Cuneiform (Uruk IV)"|
```

```
                                        estimations.df$subcorpus == "TeDDi", ]
```

Entropy rate vs. unigram entropy

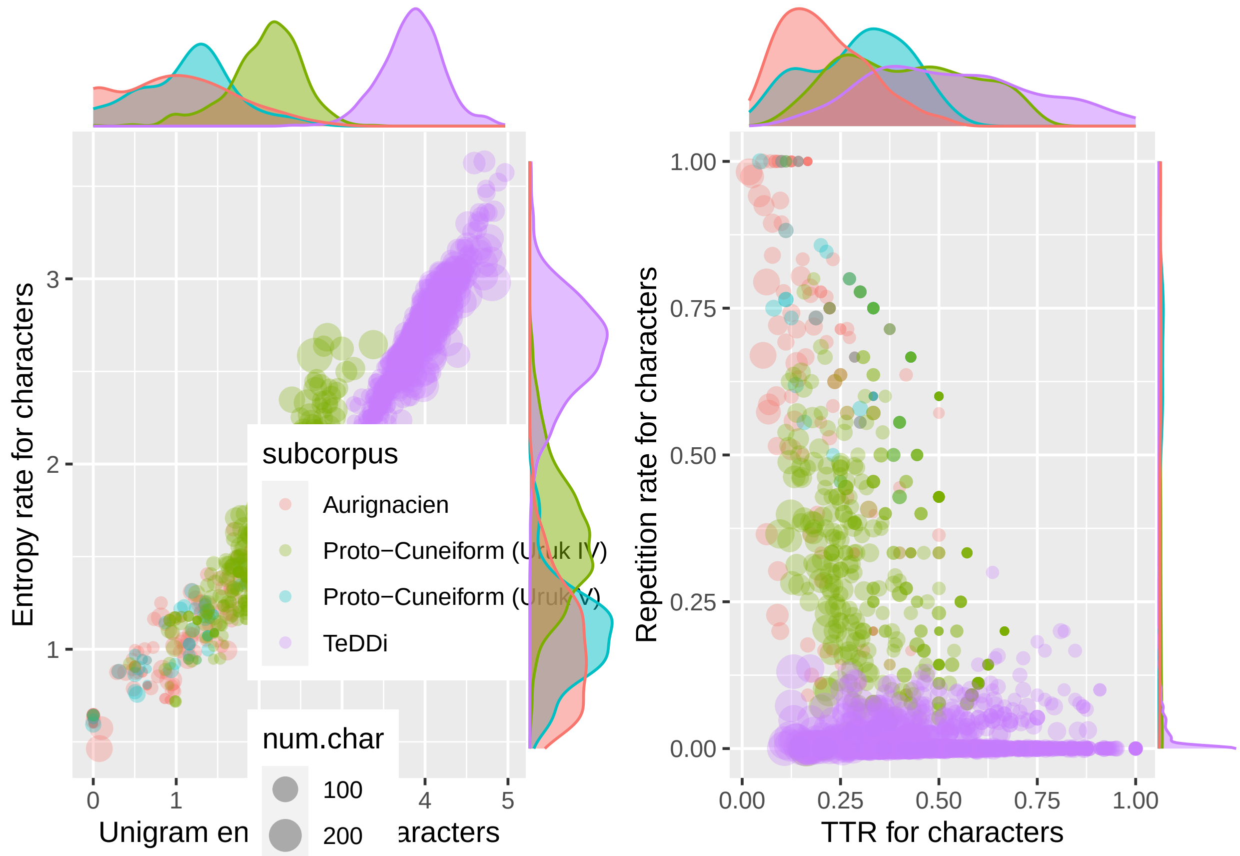
```
huni.hrate.chars.plot <- ggplot(estimations.df.selected, aes(x = huni.chars, y = hrate.chars,
                                  colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(estimations.df.selected, aes(x = ttr.chars, y = rm.chars,
                                  colour = subcorpus, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_chars.pdf",
       combined.plot, dpi = 300, width = 20, height = 10, device = cairo_pdf)
```

Swabian Jura Specific Plots

Comparisons of different caves of the Swabian Jura.

Entropy rate vs. unigram entropy

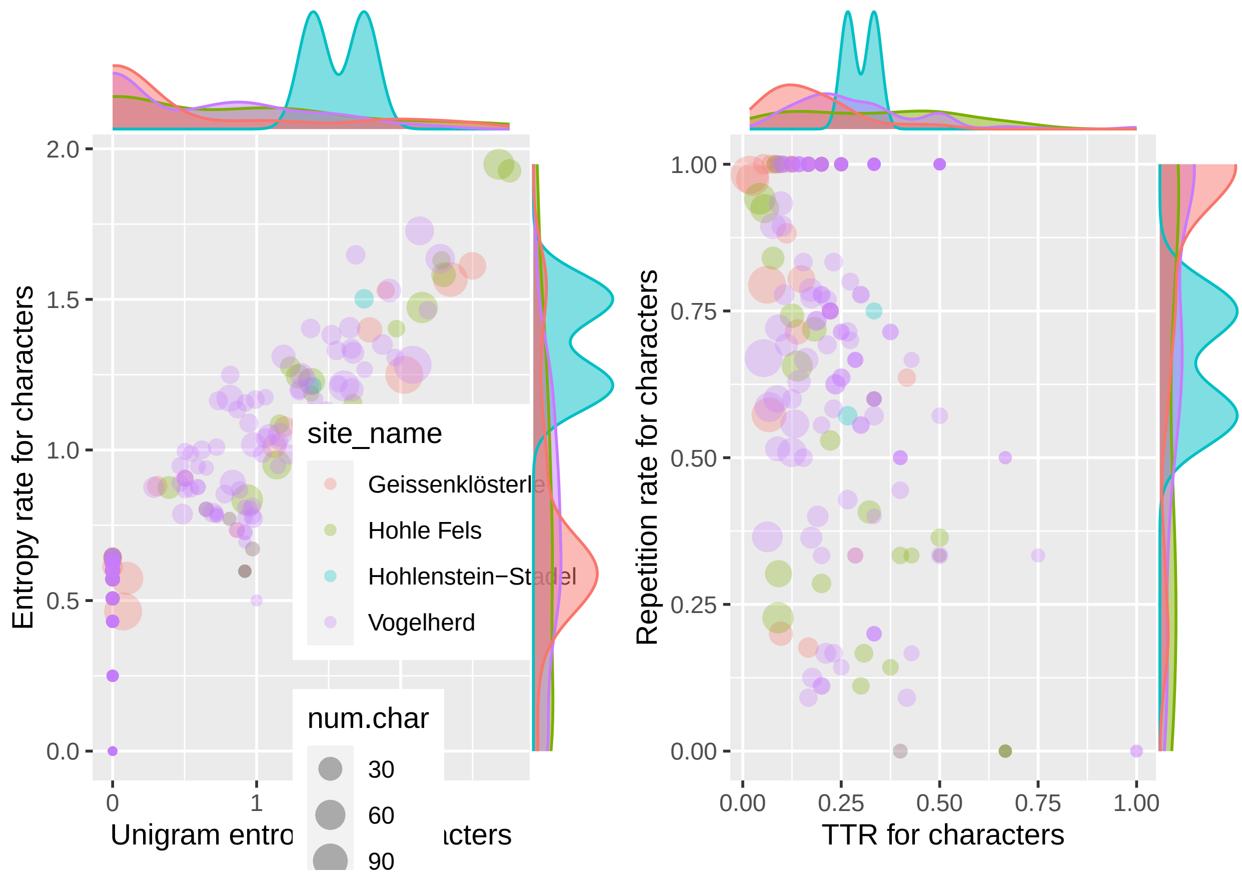
```
huni.hrate.chars.plot <- ggplot(sign.data, aes(x = huni.chars, y = hrate.chars,
                                  colour = site_name, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(sign.data, aes(x = ttr.chars, y = rm.chars,
                                           colour = site_name, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/combinedPlot_Aurignacian.pdf",
       combined.plot, dpi = 300, width = 20, height = 10, device = cairo_pdf)
```

TeDDi Specific Plots

Comparisons of different writing systems in TeDDi.

Entropy rate vs. unigram entropy

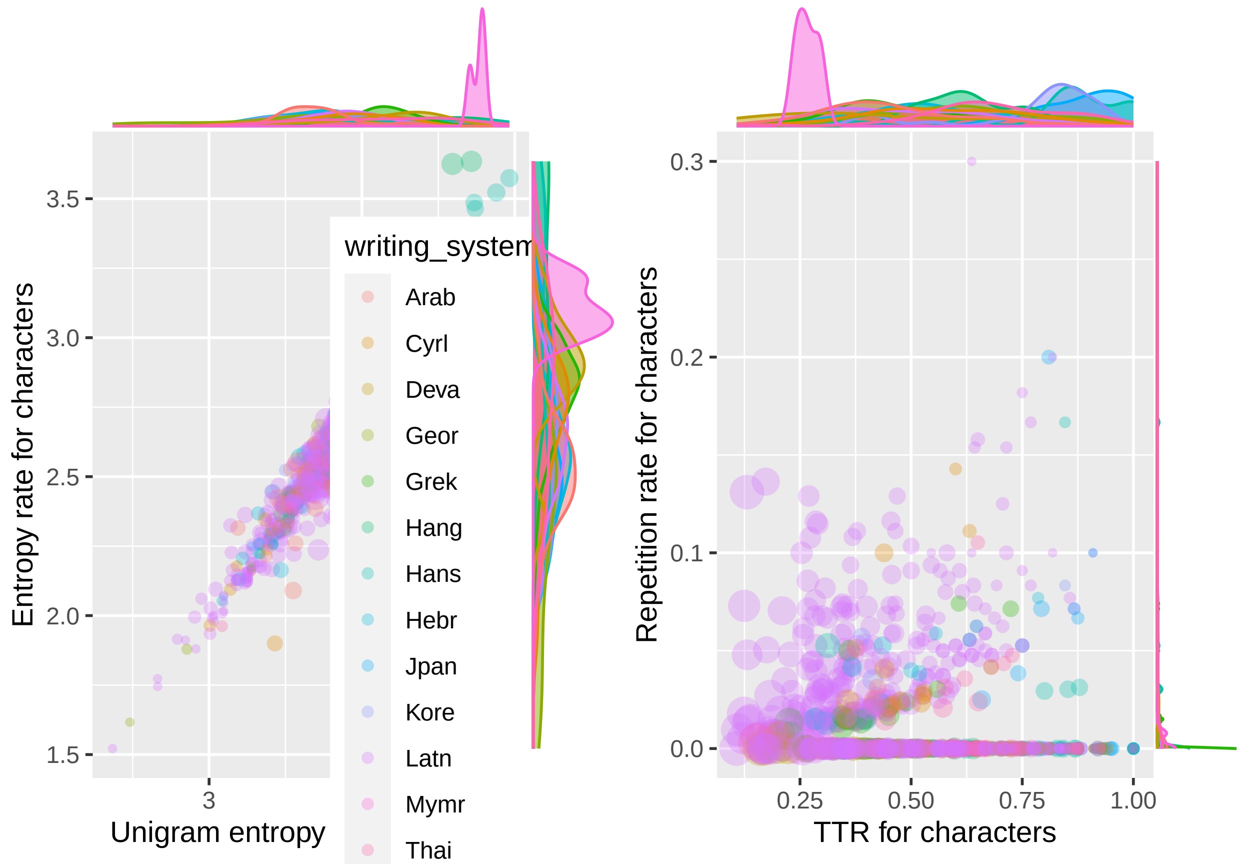
```
huni.hrate.chars.plot <- ggplot(teddi.data, aes(x = huni.chars, y = hrate.chars,
                                    colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .8) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") +
  theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#huni.hrate.chars.plot
```

TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(teddi.data, aes(x = ttr.chars, y = rm.chars,
                                    colour = writing_system, size = num.char)) +
  geom_point(alpha = 0.3) +
  #geom_density2d(alpha = .5) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") +
  theme(legend.position = "none")
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, groupFill = T, groupColour = T,
                                  type = "density")
#ttr.rm.chars.plot
```

Combine figures.

```
combined.plot <- grid.arrange(huni.hrate.chars.plot, ttr.rm.chars.plot, ncol = 2)
```



Safe complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/writingSystem_plot.pdf",
       combined.plot, dpi = 300, width = 20, height = 10, device = cairo_pdf)
```